

(d) (i) The main program:

- outputs the contents of the array using `PrintArray()`
- sorts the array using `BubbleSort()`
- outputs "Sorted"
- outputs the contents of the sorted array using `PrintArray()`

Write program code for the main program.

Save your program.

Copy and paste the program code into part **2(d)(i)** in the evidence document.

[3]

(ii) Test your program.

Take a screenshot of the output(s).

Save your program.

Copy and paste the screenshot(s) into part **2(d)(ii)** in the evidence document.

[1]

(e) The recursive function `RecursiveBinarySearch()` takes four parameters:

- an integer array
- the lower bound of the array
- the upper bound of the array
- the value to find in the array.

The recursive function performs a binary search to find the index of the value in the array.

The function returns the index of the value if it is found. The function returns `-1` if the value is not found.

Write program code for `RecursiveBinarySearch()`

Save your program.

Copy and paste the program code into part **2(e)** in the evidence document.

[6]